

PadelLens × The Lectures

Every course concept you used — and exactly where

Data Visualization for Sport · Politecnico di Milano · AY 2025/2026

For each idea: what the lecture taught, where it lives in your project, and how to answer if the professor asks.

How to read this. The course material is built almost entirely on Andy Kirk's book *"Data Visualisation: A Handbook for Data-Driven Design."* Your project deliberately follows Kirk's process, so almost every section of your work maps to a named lecture concept. The five source decks are: L1.2 Introduction to Data Visualization, L1.3 UX in Data Visualization, L2.1 Formulating the Brief, L2.2 Working with Data, and L2.3 Editorial Thinking.

Lecture 1.2 — Introduction to Data Visualization

This deck gives you the definitions and first principles. Most of it is taken directly from Kirk's handbook. Four things from here show up in your project.

Concept (lecture term)	What the lecture says	Where YOU used it in PadelLens
Definition of data visualization	"Transforming complex datasets into accessible, intuitive visual representations." Visualization is only a channel for communicating insight.	The whole premise: you turn a messy match log into accessible visual feedback. Say: "My job was to make a complex, disaggregated log readable at a glance."
The 3 principles of good dataviz: Trustworthy · Accessible · Elegant	Kirk: good viz must be (1) trustworthy (reliable, has integrity), (2) accessible (usable, unimpeded access), (3) elegant (aesthetic, refined).	TRUSTWORTHY: grounded in real Premier Padel data; honest that personal data is synthetic; no fake heatmaps (no false precision). ACCESSIBLE: WCAG-AA contrast, colour-blind-safe palette, no colour-only encoding. ELEGANT: airy 12-column layout, one typeface (Inter), minimal palette, 'annotate don't decorate'.
Aristotle's rhetoric: Ethos · Logos · Pathos	The deck cites Aristotle's three modes of persuasion: Ethos = trust/credibility; Logos = logic/facts; Pathos = emotion deliberately created.	ETHOS: real data sources + openly stated limitations. LOGOS: every chart is tied to an EDA finding — numbers carry the argument. PATHOS: the core message 'you don't have to feel your game — you can see it' and the relief of an uncluttered Home screen.
Visual encoding: Marks & Attributes (channels)	Marks = visual placeholders for each data item (points, bars, lines). Attributes = variations in a mark's properties (length, position, colour) that carry the values.	Partner bars encode win rate as LENGTH (the strongest comparison channel). Shot DNA encodes a SIGNED value with length + direction + colour. Colour is a semantic attribute (blue = neutral, red = negative). Say: "I chose the channel that fits the question."
Exploratory vs Explanatory visualization	Exploratory = you finding patterns; Explanatory = presenting a known message to an audience with focus and annotation.	Your EDA pass was exploratory; the final app is explanatory — every chart has an annotation and a single takeaway, guiding the user to the point.

Lecture 1.3 — UX in Data Visualization

This is the source for your entire UX section. Five named concepts from this deck appear in your work.

Concept (lecture term)	What the lecture says	Where YOU used it in PadelLens
User Experience (UX)	How a user interacts with and feels about a system; in dataviz it covers clarity, exploration and emotional connection (curiosity, discovery).	Your five-page, goal-oriented information architecture, designed for 'burst' use right after a match. Each page maps to a user goal, not a data type.
Usability (Learnability · Efficiency · Memorability · Errors · Satisfaction)	ISO/Nielsen components: how easily users learn, how fast they work, how well they remember, how few errors they make, and how satisfied they feel.	LEARNABILITY: padel-native words (bandeja, reves), familiar cards. EFFICIENCY: log a match in <90 s, always-visible filters. MEMORABILITY: identical nav + one accent colour. ERRORS: Set-3 gating + inline score validation. SATISFACTION: airy, encouraging tone.
Task Analysis (user goals & target users, scenarios)	Identify user goals and target users, then break the work into tasks each component supports; cognitive task analysis and usage scenarios.	Your six tasks T1–T6 ARE the task analysis. Your TARGET USERS are represented as personas — Marco (club player) and Luca (coach). Scenario: 'after a match, Marco opens the app to see how he's doing and what to drill.' Note: the lecture's term is 'identifying user goals and target users' — personas are simply how you represent them.
Heuristic Evaluation — Nielsen's 10	A discount usability method: judge the design against Nielsen's ten heuristics and fix what fails.	You evaluated your sketches against all ten and made two concrete changes: #5 Error prevention (Set-3 inputs gated) and #6 Recognition over recall (filters made always-visible). You also satisfy #1 visibility (live auto-result), #2 match the real world (padel vocab), #4 consistency, #8 minimalist design.
Cognitive load & pattern recognition	Good viz reduces cognitive load and lets users rely on fast pattern recognition.	'One number above the fold' on Home minimises cognitive load; bars and sparklines exploit pattern recognition so the user reads trends pre-attentively.

Lecture 2.1 — Hidden Thinking: Formulating the Brief

The lecture's key idea: visualization begins with a motivating CURIOSITY, not a data request. This is the backbone of your Brief.

Concept (lecture term)	What the lecture says	Where YOU used it in PadelLens
Motivating curiosity ('the true origin')	Visualization starts from a 'why' — a driving curiosity. Without it, a visualization lacks direction.	Your curiosity: "Is it actually true that I lose third sets / play badly with certain partners — or does it just feel that way?" Everything traces back to this question.
Establishing the vision (what / for whom / for where / with what / why)	The brief defines what you visualize, for whom, for where, with what data, and crucially why.	Brief §1 answers all five: a web app, for Marco the club player, used on his phone after matches, from his own match log, to confirm or correct his gut feelings.

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Refining curiosity & framing it as a question	Initial curiosity isn't fixed — refine it; a focused question yields a more effective visualization.	You refined the original 'padel journal' into the two-half (Pro Tour + personal) hybrid AFTER surveying the data. Your research questions are phrased as questions ('which partners work?', 'why do I fade?').

Lecture 2.2 — Hidden Thinking: Working with Data

The lecture gives a **four-step pipeline and a mantra**: “Data → Clean → Structure → Explore → Question → THEN Visualize.” Your data section is this pipeline, step for step.

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Data Acquisition	Sourcing and gathering the raw material; document how it was gathered (sources, sampling, cleaning).	Two layers: real pro data via the Padel API (with a committed CSV + Kaggle/scrape fallbacks) and a synthetic personal seed log.
Data Examination	Familiarise yourself with key properties and conditions; check completeness, missing values, inconsistencies.	You defined the schema of all three CSVs and checked types, ranges and completeness before building anything.
Data Transformation	Refining data through modification and consolidation.	Score-string parsing ('6-3' → numbers), derived columns (won, net balance, is_three_setter), joins (players ↔ matches), aggregations (rolling win rate, partner/surface groupings).
Data Exploration (EDA)	Use exploratory analysis to discover insights BEFORE visualizing.	Your three findings — partner effect (80/20), third-set fade (80/40), shot balance (+176/−170) — each became exactly one chart. Say: 'I did EDA before Figma, and each finding earned one chart.'
Embrace data issues (make the absence part of the story)	If the data isn't there, acknowledge it openly and make it part of the narrative.	No public amateur-padel dataset exists — so you use a documented synthetic seed and say so. The gap itself motivates the app.

Lecture 2.3 — Hidden Thinking: Editorial Thinking

Yes — the photojournalism metaphor (Angle, Framing, Focus) is straight from this lecture. So are Relevance and the five design layers. This is the most heavily-used deck in your project.

Concept (lecture term)	What the lecture says	Where YOU used it in PadelLens
Editorial thinking as photojournalism: ANGLE	The perspective you 'shoot' from — the single most important thing to show, written as a headline. You may need multiple angles but avoid too many.	Your angle is 'reflective self-analysis over time' — temporal, categorical and comparative. You explicitly excluded spatial/ball-tracking angles you can't support.
FRAMING	What you keep in vs leave out — which rows and columns you discard.	IN: what an amateur can log in 90 s (matches, per-set shot tallies, partner, surface, score). OUT: biometrics, ball-

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		trajectory heatmaps, opponent scouting.
FOCUS	What you make impossible to ignore — e.g. highlighting one number with colour or size.	The single headline on Home (last-10 win rate) plus one auto-generated plain-language insight; the ‘-40 pp’ tag on the fatigue chart.
Relevance: Timeliness · Interestingness · Pertinence · Sufficiency	Four factors that decide whether the information is worth showing this audience.	You scored them 3 / 3 / 3 / 2 and can justify each: actionable right after a match; confirms gut feelings; it’s the user’s own data; deliberately not exhaustive (the ‘2’).
The 5 layers of the design solution	Kirk’s five layers shaped by editorial thinking: Data Representation · Interactivity · Annotation · Colour · Composition.	DATA REPRESENTATION: the 9 chosen charts. INTERACTIVITY: filters (= re-framing), tooltips, drill-down. ANNOTATION: static callouts carry the headline. COLOUR: blue/red/grey semantic palette. COMPOSITION: 12-column grid, airy hierarchy.

Development Framework (E1.x) — name it, don’t dwell

These decks (Python for data viz, Streamlit basics, Streamlit data handling) underpin your Technical Implementation section — which you’re covering separately. If asked, the one-liner is: **“Python + Streamlit because this is a data-visualization project; Streamlit serves interactive charts in pure Python through the browser, so it’s cross-platform by default.”**

If the professor asks... point straight here

A fast lookup so you never freeze on ‘where did you use the framework?’

Concept (lecture term)	What the lecture says	Where YOU used it in PadelLens
“What is data visualization?”	Kirk: a channel to communicate insight via accessible visual representations (L1.2).	‘Turning my match log into accessible visual feedback.’
“What makes good dataviz?”	Trustworthy, Accessible, Elegant (L1.2).	Real data + honesty; accessible colour; airy minimal design.
“Where is Ethos/Logos/Pathos?”	Aristotle’s persuasion (L1.2).	Real sources (ethos), EDA-driven charts (logos), the core message (pathos).
“What are your marks and channels?”	Marks & Attributes (L1.2).	Bars encode win rate as length; Shot DNA uses length+direction+colour.
“What’s your UX process?”	UX, Usability, Task Analysis (L1.3).	5 goal-pages, T1–T6 tasks, Marco/Luca personas.
“Did you do a heuristic evaluation?”	Nielsen’s 10 (L1.3).	Two changes: Set-3 gating (#5), visible filters (#6).
“Where did the project start?”	Motivating curiosity (L2.1).	‘Is my third-set fade / partner effect real?’

Concept (lecture term)	What the lecture says	Where YOU used it in PadelLens
“Walk me through your data pipeline.”	Acquisition → Examination → Transformation → Exploration (L2.2).	API + seed → schema checks → parsing/joins → 3 EDA findings.
“What’s your angle / framing / focus?”	Editorial thinking (L2.3).	Reflective self-analysis; loggable-in-90s; one headline number.
“Is it relevant?”	Relevance 4 factors (L2.3).	Scored 3/3/3/2 with a reason for each.
“Where are the five design layers?”	L2.3.	Representation, Interactivity, Annotation, Colour, Composition — point to a chart for each.

One sentence that ties it all together (say this if cornered): “My whole project is structured around Kirk’s data-visualization process from the course — I formulate a brief from a motivating curiosity, work the data through acquisition–examination–transformation–exploration, apply editorial thinking with a clear angle, framing and focus, and build the solution across his five design layers, all judged against the three principles: trustworthy, accessible, elegant.”